Vectors

Exercise

Vectors

Take a look at the code below:

```r
# Assign grades to objects
albert_grade <- 80
karla_grade <- 95
ariel_grade <- 90
geoff_grade <- 90

# Adjust grades by 5 points
albert_grade <- albert_grade + 5
karla_grade <- karla_grade + 5
albert_grade <- albert_grade + 5
geoff_grade <- geoff_grade + 5
```

What is the value of albert_grade after this code runs?

a. 80
b. 85
c. 90
d. 95

Exercise

Vectors

Take a look at the code below:

```r
# Assign grades to objects
albert_grade <- 80
karla_grade <- 95
ariel_grade <- 90
geoff_grade <- 90

# Adjust grades by 5 points
albert_grade <- albert_grade + 5
karla_grade <- karla_grade + 5
albert_grade <- albert_grade + 5
geoff_grade <- geoff_grade + 5
```

What is the value of albert_grade after this code runs?

a. 80
b. 85
c. 90
d. 95

Imagine that we were dealing with grades from 50 students...

How likely would it be for us to make a mistake?

Vectors

Stores a sequence/array of values.

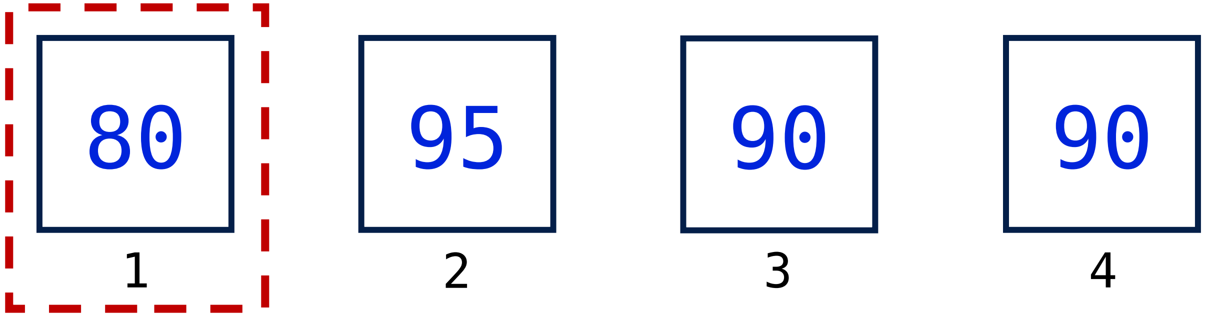
```r
grades <- c(80, 95, 90, 90)
grades <- grades + 5
```

```r
# Other type of vectors
numeric_vector <- c(1.1, 50.0, 92.562)
char_vector <- c("pizza", "soda pop", "3")
bool_vector <- c(TRUE, FALSE)
```

Vectors

Values can be **indexed** by the position

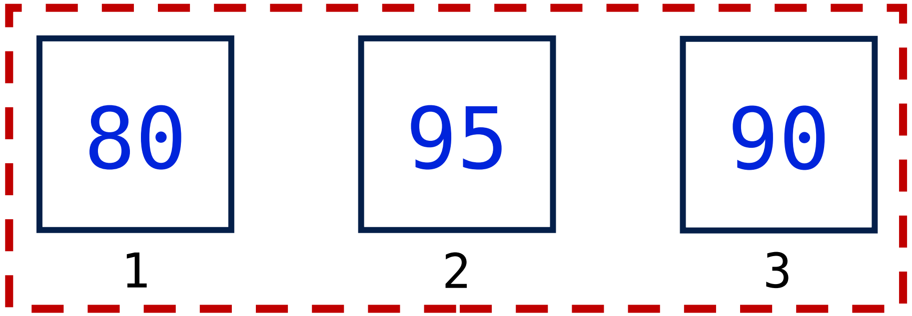
```r
grades <- c(80, 95, 90, 90)
grades[1] <- grades[1] + 5
```



Vectors

Values can be **indexed** by the position

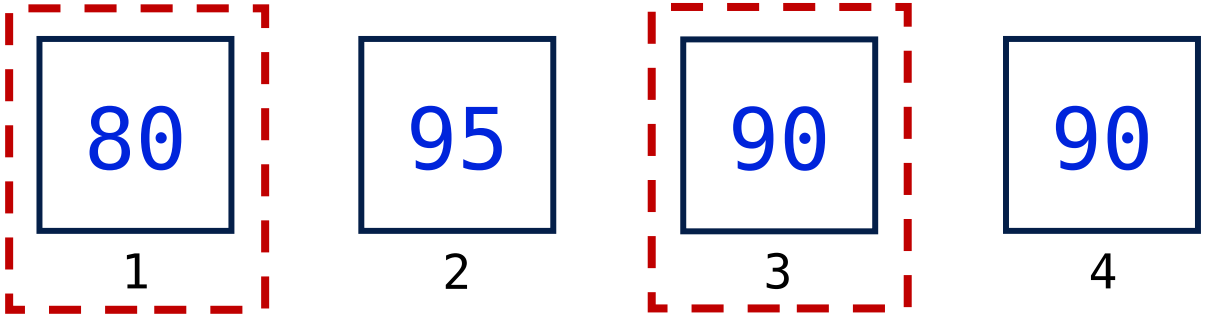
```r
grades <- c(80, 95, 90, 90)
grades[1:3] <- grades[1:3] + 5
```



Vectors

Values can be **indexed** by the position

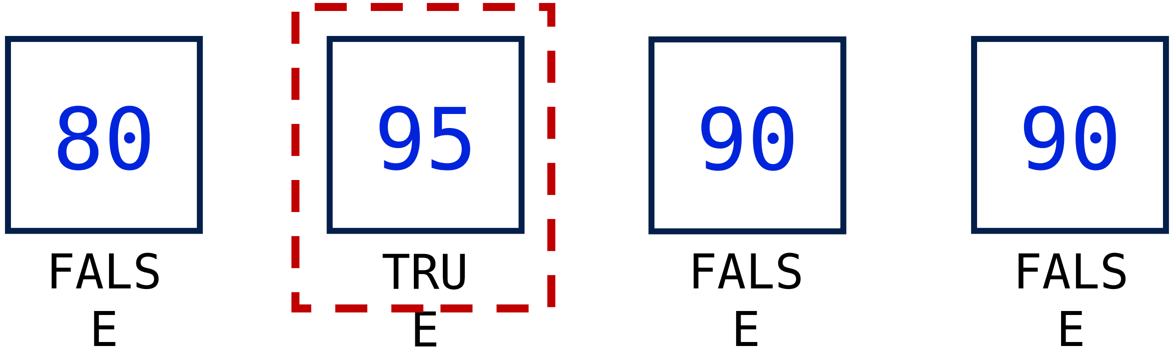
```
grades <- c(80, 95, 90, 90)
grades[c(1, 3)] <- grades[c(1, 3)] + 5
```



Vectors

Values can be **indexed** by logic

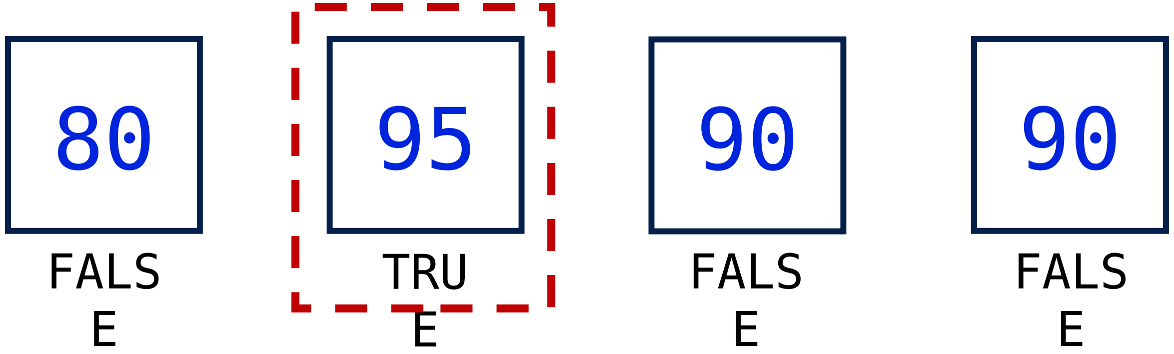
```
grades <- c(80, 95, 90, 90)
grades[grades > 92] <- grades[c(1, 3)] + 5
```



Vectors

Values can be **indexed** by logic

```
grades <- c(80, 95, 90, 90)
grades[c(FALSE, TRUE, FALSE, FALSE)]
```



Demo

Vectors

In this demo, we will:

Index vectors

Knowledge Check

Vectors

Which of the following code snippets outputs the following:

```
[1] 4 7 3
```

a)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[1:4])
```

b)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[basic_vec >= 2])
```

c)

```
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[c( 2, 3, 4 )])
```

Knowledge Check

Vectors

Which of the following code snippets outputs the following:

```
[1] 4 7 3
```

a)

```r
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[1:4])
```

b)

```r
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[basic_vec >= 2])
```

c)

```r
basic_vec <- c( 1, 4, 7, 3, 2)
print(basic_vec[c( 2, 3, 4 )])
```

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